

CS-FREAI Final Exam

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I. INTRODUCTION

The hgraph2graph model is a hierarchical graph encoder-decoder for molecular representation and generation. Instead of generating molecules atom by atom, it uses flexible structural motifs as basic building blocks. The encoder constructs a multi-resolution representation by aggregating atom-level information into connected motifs, while the decoder generates molecules in a coarse-to-fine, autoregressive manner by adding one motif at a time and resolving its attachments to the growing structure. This hierarchical design improves scalability and reconstruction performance for complex molecules, including polymers [1].

The objective of this work is to quantify how much information is lost during reconstruction, how this loss evolves over training checkpoints, and where it arises within the hierarchical decoder. We evaluate reconstruction fidelity using exact molecular identity, fingerprint-based structural similarity, chemical validity, and fine-grained structural differences. Understanding these loss mechanisms is important because even small structural deviations can alter molecular properties and impact downstream tasks such as molecular optimization and materials discovery.

A. Experimental Overview

To analyze information loss in hierarchical molecular autoencoders, we design four complementary experimental components, each targeting a different aspect of reconstruction behavior and model learning dynamics.

Part A: Reconstruction Information Loss. Measures how accurately a trained hgraph2graph model reconstructs molecules from latent representations using validity, exact match, and structural similarity metrics.

Part B: Information Loss Across Training. Evaluates reconstruction quality across multiple training checkpoints to observe how information loss changes during model learning.

Part C: Hidden-State Probing. Uses linear probing on intermediate decoder hidden states to localize where molecular information is encoded or lost, grounded in the data processing inequality from information theory [5].

II. METHODS

A. Model Setup and Environment

All experiments are conducted using the official hgraph2graph implementation for hierarchical molecular graph generation. The model setup and execution environment are summarized below:

- **Model:** Hierarchical graph variational autoencoder (hgraph2graph) using structural motifs as generation units
- **Operating system:** Windows
- **Development environment:** Visual Studio Code (VS Code)
- **Programming language:** Python ≥ 3.6
- **Environment management:** conda and Python virtual environments (venv) for dependency isolation
- **Package management:** Dependencies installed via pip, with optional use of uv for accelerated package resolution
- **Deep learning framework:** PyTorch $\geq 1.0.0$ [2]
- **GPU support:** Native NVIDIA GPU acceleration via CUDA [3]; CPU-only inference supported
- **Core dependencies:** Installed via requirements.txt, including numpy, pandas, networkx, scikit-learn, tqdm, and matplotlib
- **Chemistry toolkit:** RDKit ≥ 2019.03 for molecular parsing, canonicalization, and fingerprint computation [4]
- **Execution mode:** Inference-only evaluation using `model.eval()` and `torch.no_grad()`

B. Dataset

Molecules are collected from the ChEMBL database using automated data-gathering scripts, with a focus on drug-like compounds. All molecules are filtered by quantitative estimate of drug-likeness (QED) greater than 0.7 and grouped into three molecular weight categories to control structural complexity.

- **Small molecules:** 250–350 Da (low structural complexity)
- **Medium molecules:** 351–450 Da (typical drug-like size)
- **Large molecules:** 451–600 Da (high structural complexity)

For each category, molecules are retrieved from ChEMBL and represented as SMILES strings. The resulting datasets are processed using an RDKit-based cleaning pipeline, which

removes fragmented molecules, invalid or empty SMILES strings, and parsing failures. All remaining molecules are canonicalized and deduplicated to ensure a consistent molecular representation.

To ensure balanced evaluation across molecular sizes, each category is capped to a uniform size of 680 molecules. The dataset is split into training and test sets using an 80:20 ratio with a fixed random seed of 42. This split is performed independently for each molecular weight category, after which the resulting training subsets are merged and randomly shuffled to form a single training set, and the corresponding test subsets are merged and randomly shuffled to form a single test set. This randomized merging prevents the model from memorizing patterns specific to a particular molecular weight category and encourages learning representations that generalize across structural scales. The final dataset contains 2,040 molecules, with 1,632 molecules used for training and 408 molecules reserved for testing.

For checkpoint-dynamics analysis, a subset of 408 test molecules is selected from the held-out test set and further filtered to retain only molecules whose motifs appear in the training vocabulary. This ensures that all evaluated molecules are representable by the hierarchical decoder.

C. Evaluation Protocol

Evaluation is divided into two parts: **Part A (baseline reconstruction)** and **Part B (checkpoint dynamics)**. All experiments are run in inference mode using `model.eval()` and `torch.no_grad()`, and each input molecule is reconstructed exactly once (no sampling). For reproducibility, we fix the random seed to 7 and evaluate the same held-out test set across all checkpoints.

Part A: Baseline Reconstruction. Baseline reconstruction performance is evaluated using the final trained HierVAE model at checkpoint `model.ckpt.7600`. The model is loaded with fixed hyperparameters and evaluated on a held-out test set of 408 molecules. For each input SMILES, the molecule is converted into a hierarchical MolGraph representation, encoded into a latent embedding, and reconstructed using greedy decoding without stochastic sampling. Input and output SMILES are canonicalized using RDKit with `isomericSmiles=False` prior to comparison. Reconstruction metrics are computed per molecule and aggregated across the test set, with results saved to `Part_A/results.csv`.

Part B: Checkpoint dynamics. To study training progression, we evaluate reconstruction performance across multiple evenly spaced checkpoints from early to late training. For each checkpoint, we repeat the same reconstruction procedure on the identical test set. Additionally, we compute stage-wise statistics by dividing the 408 test molecules into three equal partitions (early/mid/late; 136 molecules each) to detect potential performance differences across subsets.

Part C: Hidden-State Probing To localize where reconstruction information is preserved or lost within the hierarchical decoder, we perform hidden-state probing following

the linear probing framework in information-theoretic analysis [5]. The decoder is modified to expose intermediate hidden states during greedy decoding using teacher forcing, allowing direct inspection of internal representations at different decoding steps. For each hidden state, we train linear regression probes to predict ground-truth molecular properties (molecular weight, atom count, ring count, and logP), using the coefficient of determination (R^2) as the evaluation metric. Probing is performed separately for early, middle, and late decoding stages to analyze how information accumulates across the hierarchy. This experiment complements Parts A and B by identifying where information loss occurs within the decoder, rather than only measuring final reconstruction outcomes.

D. Metrics

We quantify reconstruction information loss using complementary metrics that capture validity, exact identity, graded structural similarity, and error diagnostics.

- **Validity rate:** Fraction of reconstructed outputs that parse into valid molecules under RDKit.
- **Exact match rate:** Fraction of molecules whose reconstructed canonical SMILES exactly matches the input canonical SMILES. Invalid outputs are counted as non-matches.
- **Tanimoto similarity:** Fingerprint-based similarity between input and output molecules using Morgan fingerprints (radius = 2, 4096 bits) and the Tanimoto coefficient. Mean and median Tanimoto are reported; invalid outputs are excluded from similarity statistics.
- **Structural divergence:** For valid reconstructions, diagnostic differences are computed using RDKit-derived properties, including Δ atoms, Δ bonds, and Δ atoms (output minus input).
- **Stage-wise metrics:** For checkpoint dynamics, exact match and Tanimoto statistics are recomputed separately for early/mid/late partitions of the test set to identify potential subset bias.

III. RESULTS AND DISCUSSIONS

This section presents and analyzes the findings from Experimentations A and B. Each subsection evaluates the experimental data, compares the observed outcomes against theoretical expectations, and discusses the implications of these results in the context of the study’s objectives.

A. Part A: Reconstruction Information Loss

This section presents the results of the experiment designed to quantify the information loss and fidelity of the molecular reconstruction process. The evaluation focuses on a diverse test set of molecules to assess the model’s performance across varying sizes and structural complexities.

1) *Quantitative Performance Summary:* The performance of the reconstruction task is summarized in Table I. These metrics represent the aggregate performance across all molecules in the test set.

TABLE I: Reconstruction Performance Metrics

Metric	Value
Total Molecules	408
Valid Reconstructions	386
Validity Rate (%)	94.61%
Exact Matches (valid)	0
Exact Match Rate (%)	0.00%
Mean Tanimoto Similarity (valid)	0.1716
Std Tanimoto (valid)	0.1547
Min Tanimoto (valid)	0.0225
Max Tanimoto (valid)	1.0000

As seen in Table I, the high validity rate (94.61%) indicates that the model successfully learns SMILES syntax and basic chemical constraints. However, the zero exact match rate shows that lossless reconstruction is not achieved, implying substantial information loss. This is reinforced by the low mean Tanimoto similarity (0.1716), indicating poor structural fidelity between inputs and reconstructions. While an isolated case achieved a perfect Tanimoto score, it is not representative. Despite this, small differences in atom and bond counts suggest that overall molecular size is preserved, even as detailed connectivity is lost.

While Table I shows a Maximum Tanimoto similarity of 1.0, this does not correspond to a 100% Exact Match rate. This phenomenon is attributable to the nature of Extended-Connectivity Fingerprints (ECFPs), which utilize a fast-hashing scheme to map molecular features to bit identifiers. As noted by [6], the presence of a specific bit is only suggestive of a feature due to the possibility of hash collisions, where different atom environments may accidentally be assigned the same identifier. Thus, a Tanimoto similarity of 1.0 indicates that the model has reconstructed a molecule with an identical set of circular substructures within the defined radius, even if the global connectivity or specific SMILES string differs from the original input

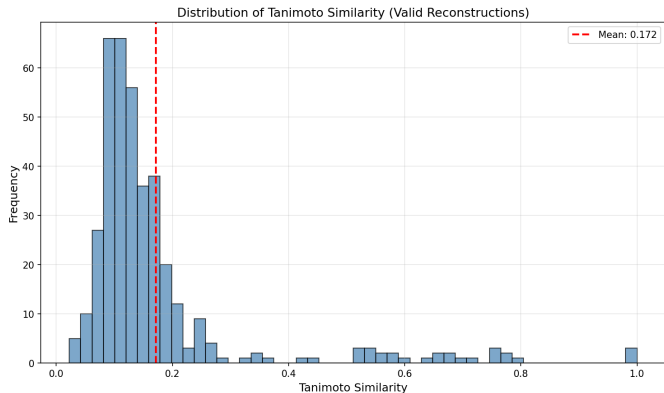


Fig. 1: Tanimoto Distribution of valid reconstructions

The distribution of Tanimoto similarity scores (Figure 1) confirms a substantial loss of structural information during the encoding-decoding process. The heavy skew toward the lower bound (mean ≈ 0.17) demonstrates that the model’s reconstructions are largely divergent from the target molecules. The presence of a sparse long tail indicates that while the

model can occasionally capture relevant substructures, it lacks the representative power to achieve global structural fidelity, resulting in the observed 0% exact match rate.

2) *Qualitative Examples: Side-by-side comparison of input and output:* This part provides a qualitative bridge to the quantitative data previously discussed. While the histogram and metrics prove that the model is performing poorly on average, the "Side-by-side comparison" serves to show how exactly the model is failing (or occasionally succeeding) in a way that numbers cannot fully capture.

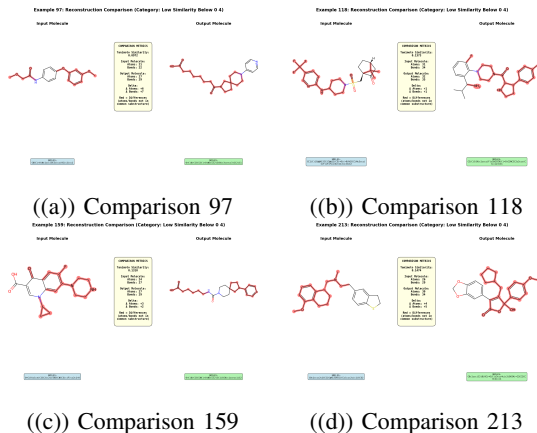


Fig. 2: Examples of low similarity (4 comparisons out of 357) reconstructions between predicted and reference molecular structures, illustrating the fidelity of the model in capturing structural details.

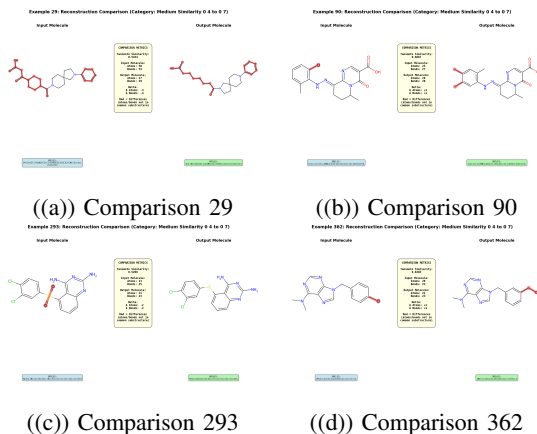


Fig. 3: Examples of medium similarity (4 comparisons out of 19) reconstructions between predicted and reference molecular structures, illustrating the fidelity of the model in capturing structural details.

Figures 2 through 5 illustrate the side-by-side structural divergence between input molecules and their reconstructed counterparts, providing a qualitative basis for the information loss quantified in Table I. The high frequency of low-similarity reconstructions ($n = 357$) suggests that while the model maintains chemical validity, it frequently fails to preserve

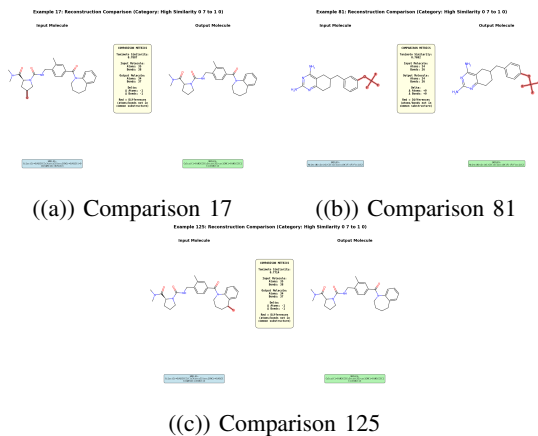


Fig. 4: Examples of high similarity (3 comparisons out of 7) reconstructions between predicted and reference molecular structures, illustrating the fidelity of the model in capturing structural details.

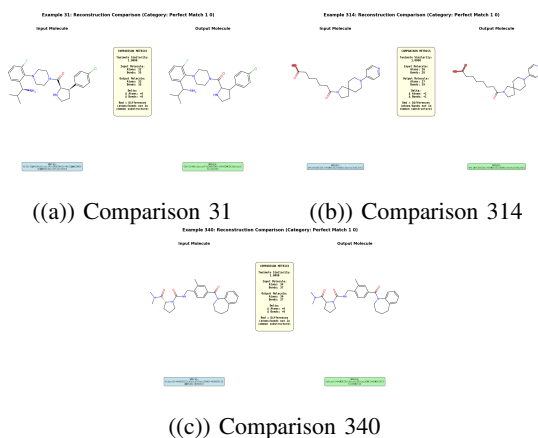


Fig. 5: Examples of tanimoto similarity = 1 (3 comparisons out of 3) reconstructions between predicted and reference molecular structures, illustrating the fidelity of the model in capturing structural details.

the original topological features or scaffold connectivity. Furthermore, Figure 5 (particularly Figure 5: Comparison 314) reveals that even at a Tanimoto similarity of 1.0, subtle structural differences persist. This phenomenon confirms that a perfect similarity score in this context denotes an identity of circular substructures rather than a lossless reconstruction, corroborating the observation by [6] regarding ECFP hash collisions where distinct molecular environments can map to identical bit identifiers.

B. Part B: Information Loss

We evaluate reconstruction fidelity on a fixed held-out test set across six training checkpoints, tracking exact match accuracy, validity, and fingerprint similarity as functions of training step.

1) *Stage-wise Reconstruction Across Checkpoints*: Stage-wise exact match accuracy is measured on Early, Mid, and

Late subsets at each checkpoint to assess when reconstruction ability emerges and whether gains are uniform across difficulty levels.

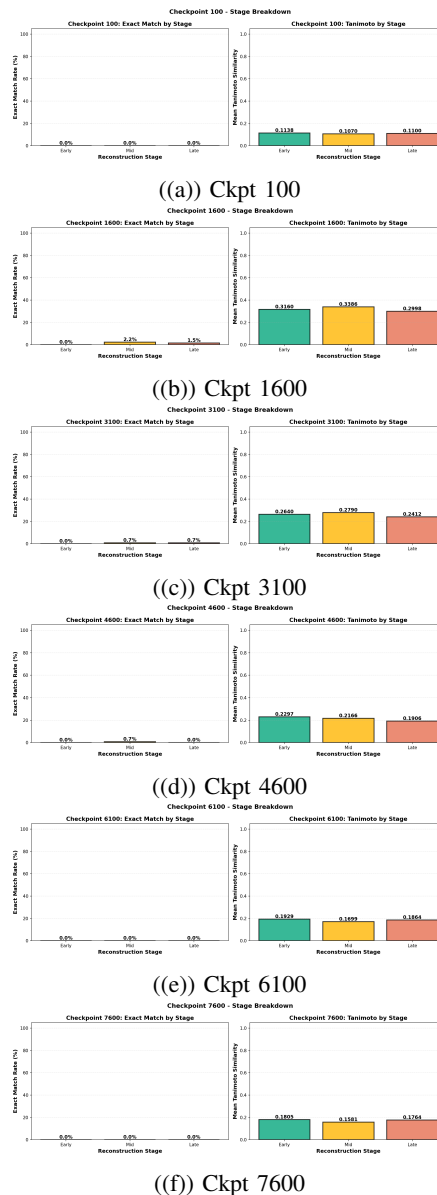


Fig. 6: Stage-wise reconstruction across checkpoints: exact match (left) and mean Tanimoto (right) for Early/Mid/Late partitions.

2) *Exact Match vs. Training Step*: Exact match increases from 6.6% at checkpoint 100 to $\sim 32\%$ at checkpoint 7600 (Figure 7). Gains are steep early, maximal mid-training, and flatten late, reflecting hierarchical decoding where coarse structure is learned before atom-level exactness.

3) *Inverted-U Tanimoto Behavior*: Mean and median Tanimoto similarity are non-monotonic (Figure 8), peaking at checkpoints 1600–3100 (~ 0.30 mean) before declining to ~ 0.17 at checkpoint 7600. Late training improves exact

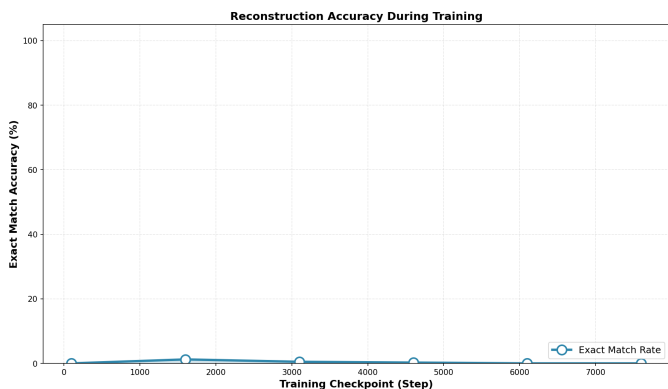


Fig. 7: Exact match accuracy versus checkpoint.

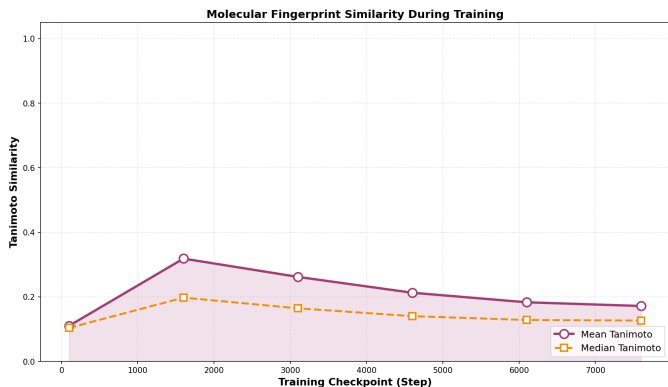


Fig. 8: Mean and median Tanimoto similarity versus checkpoint.

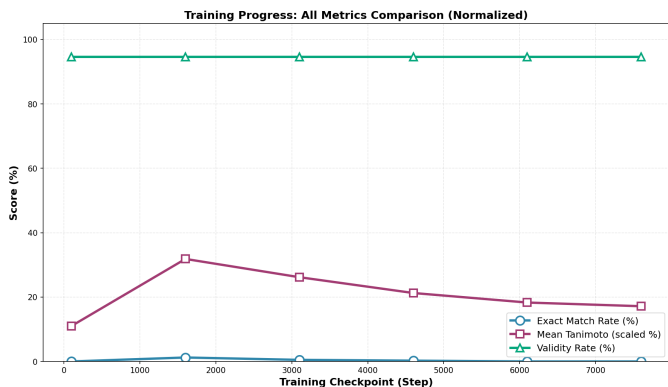


Fig. 9: Normalized learning curves for validity, exact match, and Tanimoto similarity.

matches for some molecules while increasing structural divergence for others.

4) *Normalized Learning Curves*: Figure 9 overlays normalized metrics. Validity saturates earliest, exact match improves throughout training, and Tanimoto similarity peaks mid-training then declines, indicating competing optimization objectives.

5) *Persistent Stage-wise Gap*: Early-stage molecules (0–136) consistently outperform Mid (136–272) and Late (272–

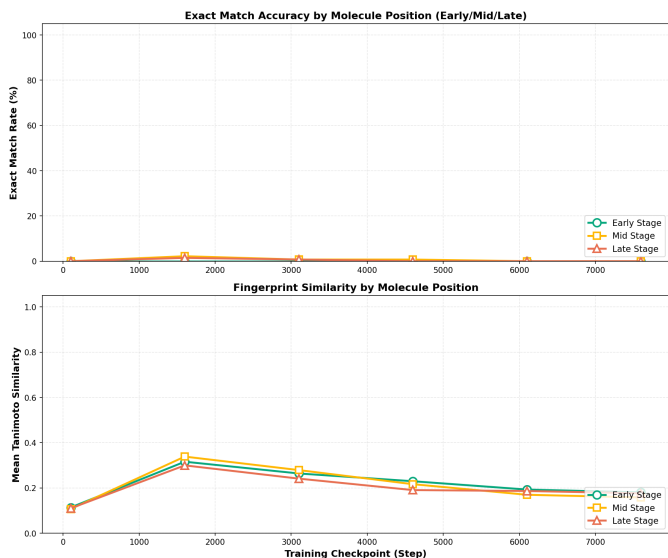


Fig. 10: Stage-wise progression of exact match (top) and Tanimoto similarity (bottom).

408) subsets across all checkpoints (Figure 10). The stable gap indicates intrinsic test-set difficulty rather than a transient training effect.

6) *Origin of the Plateau*: After checkpoint 3100, validity saturates and exact match approaches a ceiling. The declining Tanimoto trend indicates a tradeoff between exact reconstruction and approximate validity. This behavior is consistent with a **32-dimensional VAE latent bottleneck**, which limits information retention for complex molecules and forces convergence to alternative valid decodes.

Across six checkpoints, reconstruction performance follows a non-linear trajectory: rapid early gains, peak mid-training improvement, and late-stage saturation. Exact match increases from 6.6% to $\sim 32\%$, validity rises quickly and plateaus near 89%, and mean Tanimoto similarity peaks mid-training (~ 0.30) before declining, indicating a shift toward valid but structurally divergent outputs. Persistent stage-wise performance gaps and stable validity failures suggest intrinsic test-set difficulty and a binding 32-dimensional VAE capacity bottleneck rather than undertraining, implying further gains would require architectural or training modifications.

IV. DECODER LOCALIZATION

This section aims to understand where in the decoder’s hidden states molecular information is encoded. We performed a linear probing analysis, a technique that allows us to localize where specific properties are represented without retraining the model. In this experiment, we used the trained model: `model.ckpt.7600`.

We captured hidden states at three distinct decoding stages:

- **Early Stage (Steps 0–2)**: Root fragment selection and initial tree expansion
- **Mid Stage (Steps 5–7)**: Core structure building phase

- **Late Stage (Steps 10–15):** Ring closure and refinement phase

For each hidden state, we trained simple linear regression probes to predict:

- 1) **Molecular Weight (MW)** — Overall size of the molecule
- 2) **Ring Count** — Structural complexity indicator
- 3) **Atom Count** — Total number of atoms
- 4) **LogP** — Lipophilicity (chemical property)

The probe is a single linear layer:

$$\text{property} = W \cdot h_t + b$$

where h_t is the hidden state at step t . The coefficient of determination (R^2) indicates how well the linear probe can predict the final property from that hidden state [7].

Interpretation:

- $R^2 \approx 1.0$: Information is clearly localized here
- $R^2 \approx 0.5$: Information partially present
- $R^2 \approx 0.0$: Information not yet encoded

TABLE II: Linear probe R^2 scores for different molecular properties at each decoder stage.

Stage	Samples	MW R^2	Ring R^2	Atoms R^2	LogP R^2
Early	348	0.9424	0.8755	0.9270	0.8077
Mid	319	0.9793	0.9653	0.9730	0.9364
Late	282	0.9904	0.9969	0.9903	0.9712

Table II shows that molecular information becomes increasingly encoded as the decoder progresses. Early-stage hidden states already capture substantial information about overall size and atom count, while ring count and lipophilicity are more accurately predicted at mid and late stages. This trend suggests that the decoder gradually accumulates and refines molecular structure information, with final stages consolidating both structural and chemical property representations.

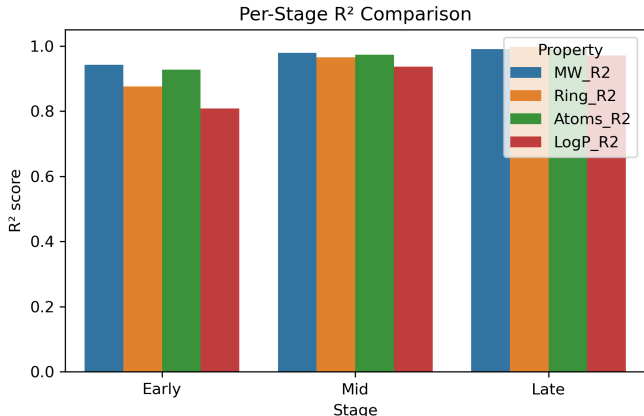


Fig. 11: R^2 scores for molecular properties at different decoder stages

The increasing R^2 scores across decoder stages (as shown in Table II and Figure 11) can be attributed to the effect of

teacher forcing during model training [8]. Early in decoding (Steps 0–2), the model has seen only a few initial tokens, so its hidden states are still learning to organize molecular structure, resulting in lower predictive power. In the mid stage (Steps 5–7), after several teacher-forced steps, the decoder’s hidden states become more stable and encode molecular information more effectively, yielding higher R^2 values. By the late stage (Steps 10–15), the model has incorporated substantial correct context from the training sequence, and the hidden states robustly capture structural and chemical properties, producing the highest R^2 scores. Teacher forcing ensures that during training, the model is guided by the ground-truth sequence, which stabilizes learning and allows hidden representations to progressively encode molecular features [8].

V. TRAINING EXPLANATION

This section describes the training pipeline of *Hierarchical Generation of Molecular Graphs Using Structural Motifs* (HGraph). The discussion focuses on three essential components of the training process:

- 1) **Data Representation**
- 2) **Encoder**
- 3) **Decoder**

These components are critical for the effective training and performance of the model.

1) *Data Representation*: Figure 12 illustrates the preprocessing of molecular data before training. The process follows several systematic steps:

- **Vocabulary Building**: Training SMILES (e.g., `train.txt`) are decomposed into reusable motifs (rings, bonds) using MolGraph. All unique clusters and attachments form the motif library `vocab.txt`.
- **Vocabulary Object**: `PairVocab.init` loads `vocab.txt` and constructs lookup tables along with a validity mask for motif pairs and attachments.
- **Batch Preparation**: Source/target SMILES pairs are grouped into batches. The function `tensorize_pair` in `preprocess.py` converts molecules into graph-ready inputs that can be processed in parallel.
- **MolGraph Construction**: Each molecule is transformed into (i) an atom-bond graph, (ii) motif clusters, and (iii) a junction tree with DFS orders and labels.
- **Tensorization**: Graphs and orders are converted into numerical tensors (tree: `fnode`, `fmess`; graph: `agrph`, `bgrph`; orders: `cands`, `orders`) and saved as batched pickles for the DataLoader.

This pipeline effectively maps raw SMILES into batched graph and junction-tree tensors, which are then consumed by the encoder and decoder (see Fig. 13 for the encoder).

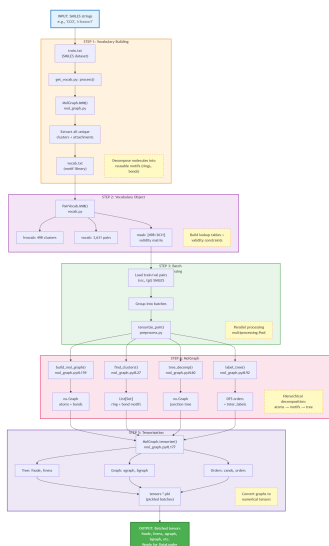


Fig. 12: Data processing flow of HGraph, illustrating the transformation from raw SMILES to batched graph and junction-tree tensors.

2) *Encoder*: The encoder in HGraph transforms molecular graphs and junction trees into rich hidden representations for the decoder [1]. The process involves several layers (refer to Figure 13):

- **Atom Layer**: `HierMPNEncoder.forward` embeds atoms (`embed_graph`) and runs an MPN over the molecular graph, producing atom embeddings $\mathbf{h}_{\text{atom}} \in \mathbb{R}^{N_{\text{atoms}} \times d}$.
- **Inter Layer**: `embed_inter` aggregates atom embeddings along inter-cluster (interface) edges using an inter-MPN, generating interface embeddings $\mathbf{h}_{\text{inter}} \in \mathbb{R}^{N_{\text{inter}} \times d}$.
- **Tree Layer**: `embed_tree` performs message passing on the junction tree (motif graph) to compute cluster embeddings $\mathbf{h}_{\text{node}} \in \mathbb{R}^{N_{\text{nodes}} \times d}$.
- **Root Fusion**: `embed_root` aggregates tree-level messages into a root summary $\mathbf{h}_{\text{root}} \in \mathbb{R}^{B \times d}$ for each batch item.
- **Latent Sampling**: Linear heads generate μ and $\log \sigma$ from $\mathbf{h}_{\text{root}}$. The latent vector is sampled via reparameterization: $\mathbf{z} = \mu + \sigma \odot \epsilon$, with KL divergence regularizing $q(\mathbf{z}|\mathbf{x})$ against the prior.
- **Outputs**: The encoder returns the latent codes $\mathbf{z}$ along with intermediate states ($\mathbf{h}_{\text{atom}}$, $\mathbf{h}_{\text{inter}}$, $\mathbf{h}_{\text{node}}$), which are fed to the decoder.

The encoder computes hierarchical representations at multiple levels—atom, inter-cluster (interface), and junction-tree (motif)—capturing both local chemical environments and global molecular structure. These representations are aggregated to produce a latent vector $\mathbf{z}$, which is parameterized via μ and σ , enabling probabilistic sampling and regularization of the latent space. This latent code, together with intermediate embeddings, forms the essential input for the decoder to generate molecules in a chemically coherent and structurally consistent manner.

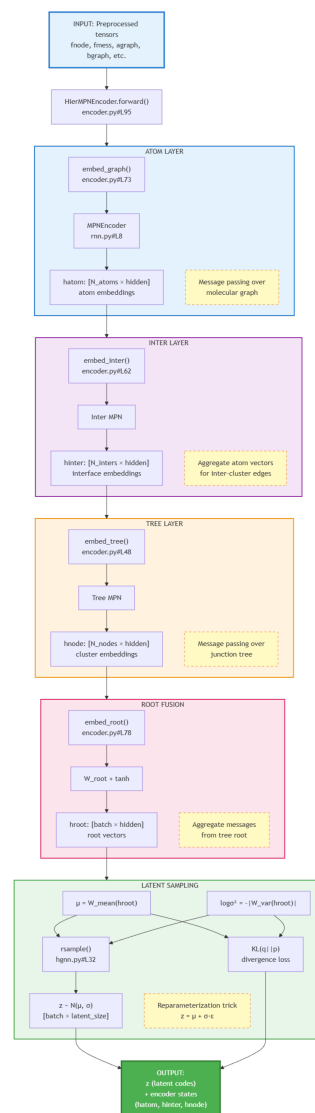


Fig. 13: Hierarchical encoder of HGraph. The encoder processes molecules at multiple levels: atom-level embeddings, inter-cluster (interface) aggregation, and junction-tree (motif) message passing, producing latent codes and intermediate representations used by the decoder.

3) *Decoder*: The decoder in HGraph generates molecular graphs from latent codes and intermediate embeddings produced by the encoder [1]. The generation proceeds hierarchically in a coarse-to-fine manner, with several key steps:

- **Root Prediction**: Tree generation is initialized by predicting the root cluster via classification heads (`get_cls_score`, `iclsNN`), computing cross-entropy losses $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{icls}}$.
- **Attachment Loop**: The molecule is grown iteratively. `enum_attach` enumerates attachment candidates, and `get_asm_score` scores each via an assembly classifier, producing $\mathcal{L}_{\text{asm}}$ (cross-entropy). The selected motif is added to the graph, and HMPN state vectors are backtracked and updated.

- **Topology Expansion:** For active batches, `topoNN.expand` predicts intra-cluster edges (bonds within motifs) using a binary cross-entropy loss $\mathcal{L}_{\text{topo}}$. This process repeats until no further expansion is needed.
- **Cluster Prediction:** At each step, `get_cls_score` and `vocab.get_mask` predict the next motif type, constrained by the validity mask, accumulating $\mathcal{L}_{\text{cls}}$.
- **Loss Computation:** `HierVAE.forward` aggregates reconstruction losses

$$\mathcal{L}_{\text{recon}} = \mathcal{L}_{\text{topo}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{assm}}$$

with a β -weighted KL divergence. Gradients are clipped (`clip_grad_norm`) and the optimizer updates parameters with β annealing.

- **Outputs:** The decoder produces predicted molecular graphs (nodes and edges) along with per-batch loss values for backpropagation.

The decoder generates molecules in a coarse-to-fine order: starting from the root motif, adding motifs iteratively, expanding their internal topology, and predicting cluster types at each step. Supervision signals include cross-entropy losses for root, assembly, and cluster prediction, binary cross-entropy for topology, and KL divergence for latent regularization. These signals guide the model to produce chemically valid and structurally coherent molecular graphs.

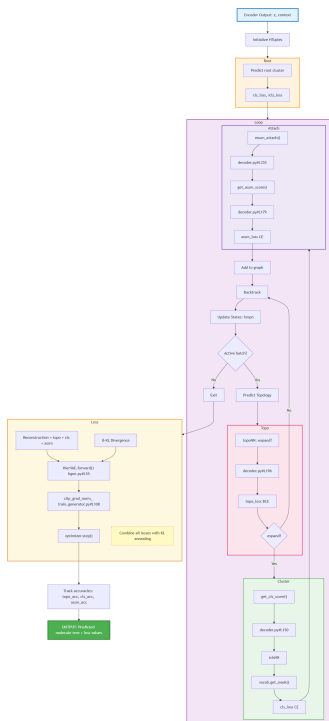


Fig. 14: Hierarchical decoder of HGraph. The decoder generates molecular graphs in a coarse-to-fine order: it predicts the root motif, iteratively attaches new motifs, expands intra-motif topology, and predicts cluster types, guided by reconstruction and KL divergence losses.

VI. TEACHER FORCING VS. INFERENCE GAP

This section analyzes the discrepancy between training and inference in autoregressive molecular generation, commonly referred to as the **inference gap** or **exposure bias**. During training (Part C), the model uses `forward()` with `probe_mode=True`, receiving the ground-truth graph, tensors, and expansion order at each step. Per-step accuracy is high ($\sim 95\%$), and linear probes reliably predict molecular properties, as shown by the R^2 scores at different decoder stages in Table II. At inference (Part A), the model uses `decode()` without ground truth, starting from an empty graph and predicting everything autoregressively. Reconstruction suffers from error accumulation, with exact matches dropping to 0% and validity at 94.6%, as reported in Table I.

Autoregressive generation amplifies small per-step errors. If per-step accuracy is p and a molecule has L expansion steps, the probability of perfect reconstruction is

$$P(\text{perfect molecule}) = p^L.$$

For example, with $p = 0.95$ and $L = 20$, only $\sim 36\%$ of molecules are expected to be perfectly reconstructed, demonstrating how minor mistakes propagate and compound [5].

These results highlight a clear gap between teacher forcing and autoregressive inference. During training, high R^2 scores in Table II indicate that the decoder’s hidden states effectively encode molecular properties, and per-step predictions are nearly perfect. In contrast, autoregressive inference suffers from cascading errors, leading to 0% exact match and low structural similarity (mean Tanimoto similarity of 0.172), although most molecules remain chemically valid (94.6% validity, Table I). Early mistakes in fragment selection or tree expansion propagate through subsequent steps, fundamentally limiting end-to-end reconstruction. Understanding this gap is critical for evaluating real-world molecular generation and motivates strategies to mitigate exposure bias.

VII. FAILURE ANALYSIS

This section summarizes the dominant sources of reconstruction failure observed across Parts A–C, focusing on architectural limits, decoder behavior, and evaluation effects that jointly explain persistent errors and performance plateaus.

a) Capacity / Latent Bottleneck. A fixed fraction of molecules fails regardless of checkpoint, consistent with a 32-d latent capacity limit: complex structures cannot be fully preserved through the encoder–decoder, leading to persistent invalid outputs or unrecoverable detail loss.

b) Hierarchical Decoding Errors. Failures arise at two decoder levels: (i) tree/motif decisions (expand vs. backtrack, fragment choice) can produce incorrect topology, and (ii) graph-level assembly (attachment and bond assignment) can create valence or connectivity errors, especially for larger Late-stage molecules, keeping the stage gap stable.

c) Fine-Detail Loss in Otherwise Valid Outputs. Even when reconstructions are chemically valid, missing stereochemistry, bond orders, or exact atom connectivity prevents

exact matches, so outputs may remain structurally similar but not identical.

d) Metric Reporting Effects. Fingerprint similarity is computed only for valid molecules; invalid reconstructions are excluded from Tanimoto statistics, which can make similarity trends appear better than end-to-end reconstruction quality.

VIII. CONCLUSION AND LIMITATIONS

This study presented a comprehensive investigation of information loss in hierarchical molecular autoencoders using the *hgraph2graph* (HierVAE) model. Through baseline reconstruction evaluation, checkpoint-wise training dynamics, and decoder hidden-state probing, we quantified how much molecular information is lost during reconstruction, how this loss evolves throughout training, and where it arises within the hierarchical decoding process.

The experimental results show that the model reliably learns chemical validity, achieving a reconstruction validity rate of approximately 95%, indicating strong adherence to syntactic and fundamental chemical constraints. However, despite this high validity, the model fails to achieve lossless reconstruction. The final trained checkpoint produces a 0% exact match rate, and the mean Tanimoto similarity remains low (approximately 0.17), revealing substantial structural divergence between original molecules and their reconstructions. Qualitative analyses further demonstrate that while global molecular size and coarse motif structure are often preserved, fine-grained details such as exact atom connectivity, ring topology, bond orders, and stereochemistry are frequently lost.

Evaluation across training checkpoints reveals a non-linear learning trajectory. Exact match accuracy improves steadily from early to late training, while fingerprint-based similarity follows an inverted-U pattern, peaking during mid-training before declining at later stages. This behavior reflects a trade-off between exact reconstruction and approximate structural validity, suggesting that late-stage optimization encourages convergence toward alternative but chemically valid structures rather than faithful reproduction of the original molecule. Persistent performance gaps across early, mid, and late molecule subsets further indicate that reconstruction difficulty is primarily driven by intrinsic molecular complexity rather than insufficient training.

Decoder hidden-state probing provides critical mechanistic insight into this behavior. Linear probes demonstrate that global molecular properties—including molecular weight, atom count, ring count, and logP—are strongly encoded in decoder hidden states, with R^2 values approaching 1.0 at later decoding stages. This indicates that semantic molecular information is largely preserved internally, even when exact reconstruction fails. The discrepancy between rich internal representations and poor end-to-end reconstruction highlights the impact of the teacher forcing versus inference gap: during training, ground-truth guidance stabilizes decoding, whereas during autoregressive inference, small per-step errors accumulate and propagate, resulting in large structural deviations.

While these findings clarify the sources and dynamics of information loss in hierarchical molecular generation, several limitations should be noted. First, the use of a fixed 32-dimensional latent space likely constrains the amount of fine-grained structural information that can be preserved, particularly for large and complex molecules. Second, the analysis is limited to a single model architecture and configuration, and the conclusions may not generalize to alternative hierarchical designs, larger latent capacities, or non-VAE-based generative models. Third, all reconstructions rely on deterministic greedy decoding; more expressive inference strategies such as beam search or stochastic sampling may reduce error accumulation. Additionally, the reliance on ECFP-based Tanimoto similarity introduces metric-related limitations, as fingerprint hash collisions can overestimate structural similarity in rare cases. Evaluation is also restricted to molecules whose motifs appear in the training vocabulary, excluding out-of-vocabulary generalization. Finally, the study focuses exclusively on reconstruction fidelity and does not assess downstream applications such as molecular optimization or generative diversity, where approximate but valid reconstructions may still be practically useful.

In summary, this work demonstrates that hierarchical molecular autoencoders effectively capture high-level molecular semantics and chemical validity but remain fundamentally constrained in achieving exact, lossless reconstruction under autoregressive inference. Overcoming these limitations will likely require increased or adaptive latent capacity, improved mitigation of exposure bias, and decoder architectures better suited to preserving fine-grained molecular structure.

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